

TwinMutes: Exploring Cooperative Behaviors of Two Pololu 3Pi+ Robots in Object Manipulation Scenarios

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Abstract—This paper investigates the impact of robot orientation on the collaborative performance of two Pololu 3Pi+ robots for manipulation tasks, such as object pushing, in constrained environments. The paper explores how the robots' orientation affect the performance of the robots in pushing an object for a set distance.

The paper examines two configurations: 1) Robots face the box from the same side and push together 2) Robots face each other with the box in between. The paper describes the implementation of the collaborative effort of robots using bump sensors. It describes the results of the experiment, which support the hypotheses and show that Configuration-2 exhibits greater stability compared to Configuration-1.

I. INTRODUCTION

Collaborative robotics has become a burgeoning field with applications ranging from industrial automation to service- oriented tasks. The advent of robot platforms such as Pololu 3Pi+ robots, has opened avenues for exploring cooperative behaviors in constrained environments. This experiment delves into the collaborative capabilities of Pololu 3pi+ robots in the context of object pushing, a fundamental scenario with implications for tasks like warehouse logistics and object manipulation.

This study explores how object mass and robot orientation affect the performance of two 3pi+ robots handling a object. The experiment examines two configurations:

- 1) Robots face the object from the same side and push together (Fig-1)
- 2) Robots face each other with the object in between (Fig-2)

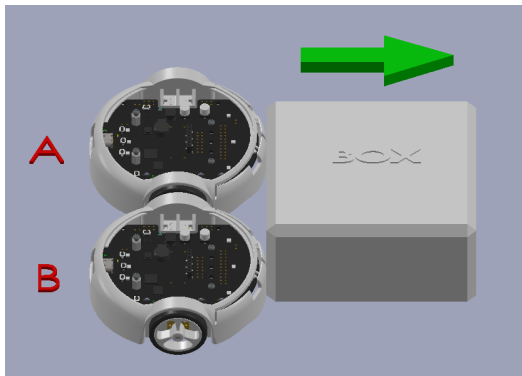


Fig. 1. Robots A (Forward) and B (Forward) driving the object in the direction of the arrow (Configuration- 1)

The fundamental motivation behind this experiment is to understand the impact of robot orientation on the effectiveness of two robot manipulation tasks. Understanding

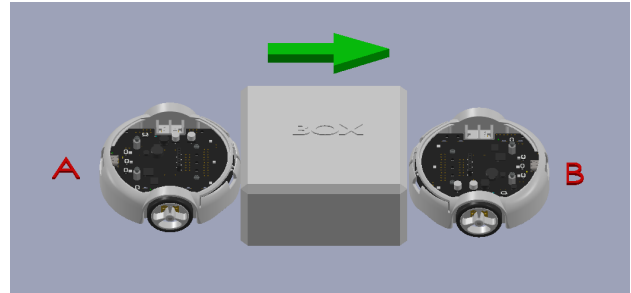


Fig. 2. Robots A (Forward) and B (Backward) driving the object in the direction of the arrow Configuration- 2

and predicting the implications of orientation is key to developing efficient and resilient manipulation algorithms.

For instance, a study by Chen et al. [1] proposed a cooperative pushing algorithm for two robots to move a heavy object. The algorithm utilizes a shared control strategy that enables the robots to adjust their pushing forces based on the object's weight and the desired movement trajectory. Another study by Kwon et al. [2] investigated the effects of robot orientation on the stability of two-robot manipulation. The results showed that the stability of the manipulated object is influenced by the relative orientation of the robots.

These studies provide valuable insights into two-robot manipulation, but further research is needed to comprehensively understand the impact of various factors, such as object weight and robot configuration, on the performance of two-robot manipulation tasks.

This research has direct implications for real-world applications of two-robot manipulation. For example, in warehouse environments, two robots can work together to transport heavy objects more efficiently than a single robot. The findings of this study can be used to optimize the coordination of these robots to ensure efficient and stable manipulation.

In addition, this research is relevant to the development of robotic systems for hazardous environments. Two robots can work together to enter and manipulate objects in hazardous environments that are too dangerous for a single robot. The results of this study can be used to design algorithms that ensure the safety and stability of robots working in such environments.

A. Hypothesis Statement

The proposed robot-based object-pushing system is hypothesized to effectively transport objects towards designated locations with high precision and accuracy, without

the use of wireless communication but by synchronised coordination of objects using bump sensors.

Hypothesis 1:

The time required for the robots to effectively manipulate the object to a specified distance is shorter with Configuration- 1 than with Configuration- 2. (H1)

This hypothesis is based on the assumption that Configuration- 1 will enable the robots to apply more coordinated and synchronised force, leading to faster pushing as they are on the same side of the object due to the parallel effort.

Hypothesis 2:

The frequency of robot misalignment during the manipulation process will exhibit that Configuration- 2 is more stable. (H2)

This hypothesis proposes opposite orientation in configuration 2 will help in counteract any deviations from desired position thereby giving more stability and effectiveness in manipulating the object.

II. IMPLEMENTATION

The experiment was conducted using two 3pi+ robots, which are small differential drive robots equipped with two bump sensors, two motors, controlled by an ATMEL ATMEGA32u4 Microcontroller each[3].

Pulse width Modulation (PWM) is a technique that varies the power output of a digital signal by turning it on and off very quickly. The ratio of on-time to off-time is called the duty cycle, and it determines the average voltage and current delivered to the motor. By changing the duty cycle, the speed of the motor can be adjusted.

The PWM is adjusted by a weighted measurement method through a feedback control loop (see Equation 1-3), for controller we used the proportional gain to respond smoothly to changes in the sensor readings of the robot, which is then fed into the motor controller to adjust the speed and direction of each motor of the robot. Thereby, contributing in adjusting the orientation of the robot with respect to the object. Feedback control is proportional gain control, which adjusts the output by a factor that is proportional to the error. This ensures that the output responds quickly and smoothly to changes in the input or the environment.

Weighted Measures:

$$W_{left} = Normalized_{LS} + (Normalized_{RS} \times 0.5) \quad (1)$$

$$W_{right} = Normalized_{RS} + (Normalized_{LS} \times 0.5) \quad (2)$$

Control Signal with Proportional Gain:

$$ControlSignal = (W_{left} - W_{right}) \times K_p \quad (3)$$

Where, LS is left bump sensor reading, RS is right bump sensor reading, w_{left} is weighted left bump sensor, w_{right} is weighted right bump sensor, and K_p is proportional gain.

The state machine (Fig. 5) provides a better understanding of the working of the algorithm. The state machine of the robot consists of four states: idle, forward, adjust left, and adjust right. For Configuration- 2 an additional state:

backward is also added. The transitions between the states are triggered by the bump sensor readings. From the initial state of idle, robots moves forward until the bump sensors are activated when the object is detected, then the robots transition to the forward state.

There are two separate controllers used for the experiments based on the operations described as follows:

- 1) In Configuration- 1, both robots are oriented facing the same side of the object and require same controller for their respective operations as they have mutual objective i.e., to push the object along a certain distance and conditions.
- 2) In Configuration- 2, where both the robots face each other along the opposite sides of the object, the forward robot has the controller same as that of Configuration- 1 whereas the backward robot requires a modified controller as the method for readjustment with a backward bias. For example, if the right sensor touches the object, it means the robot is too far to the left. So the robot moves the right wheel backward to face the object again.

It is important to note that the backward robot does not exert any effort into pushing the object. The role of the backward robot is to support the orientation of the object to the centre of the forward robot, in order to keep the forward robot from misalignment. In this report, a misalignment is defined as a deviation of the centre of the robot's bump sensor from its alignment with the object.

III. EXPERIMENT METHODOLOGY

A. Overview of Method

A rectangular object was used in the experiment which had a mass of 244 grams with dimensions of 20 x 13.5 x 8 cm. The bump sensors were calibrated manually by keeping the object in the front (centre, left and right) and away from the robot. Then, the readings were recorded as thresholds to ensure consistent object detection in the controller. Both robots were positioned on a flat surface, facing the object on the length side (Configuration- 1) or facing each other with the object on the length side keeping the object in the middle (Configuration- 2). Base PWM values of 60 for forward bias and 55 for backward bias (Configuration- 2 only) in order maintain contact with the object. The reason backward bias is slightly less than the forward bias is to ensure that the backward robot provides minimal resistance to the forward robot.

During start of each trial, the robots were placed in the orientation as shown in Fig. 3 and Fig. 4 for Configuration- 1 and Configuration- 2 respectively. It was ensured that the distance between the centre of the two robots in Configuration- 1 was 11 cm. The robots were started at the same time from a distance of 1 cm from the object for both configurations and moved forward until the robots' bump sensors were triggered when they detected the object. Using this bump sensor data, the robots calculated the necessary weighted measurement of the PWM values to drive the motors and push the object.

The robots adjusted their positions to align the object between their two bump sensors by adjusting towards right or left. The serial data for misalignment with corresponding elapsed time was logged and captured for data analysis.



Fig. 3. Configuration- 1



Fig. 4. Configuration- 2

The experiment monitored the ability of the robots to successfully push the object for a constant distance of 80 cm. The distance is based on the availability of a controlled environment (section III B).

The data collection process involved recording the following information for each trial: object weight (constant, based on the robots capacity to move the object), time taken for robots to lose contact with the object, and instances of robot misalignment. The data was analyzed to determine the effects of PWM values and the orientation of the robots' on object pushing performance.

B. Discussion of Variables

Independent Variable:

Orientation:

- The configuration of the robots (Configuration- 1 or Configuration- 2) and the base PWM values (60 or 55) for the forward and backward robots.

Object Position:

- Consistent initial position of the object relative to the robots to eliminate confounding factors

Dependent Variable:

Time Taken for Robots to complete the target distance

- Measures elapsed time corresponding to misalignment count.
- Shorter time indicates more effective pushing

Instances of Robot Misalignment

- Indicates errors in robot positioning and coordination

- A higher misalignment rate suggests more challenging conditions for maintaining proper alignment

Controlled Variable(s):

Mass and dimensions of the object:

- Mass of the object remains constant at 244 gms throughout the experiment across both configurations.
- Dimensions of the object remains constant at 20 x 13.5 X 8 cm throughout the experiment across both configurations.

Robot Configuration (Configuration- 1 or Configuration- 2)

- Identical arrangement of robots relative to the object to isolate PWM effects

Environmental Factors

- Impact of battery life on the robots' performance is controlled by having each trial with full battery capacity.
- Factors such as friction of surface and inertia of the object are disregarded in this experiment.
- Constant environment for all trials to neglect the influence of temperature, humidity, or external disturbances.

By meticulously controlling these variables, the experiment isolates the effects of the PWM value (the independent variable) on the dependent variables, providing valuable insights into the performance of two robots manipulating a object in a controlled environment.

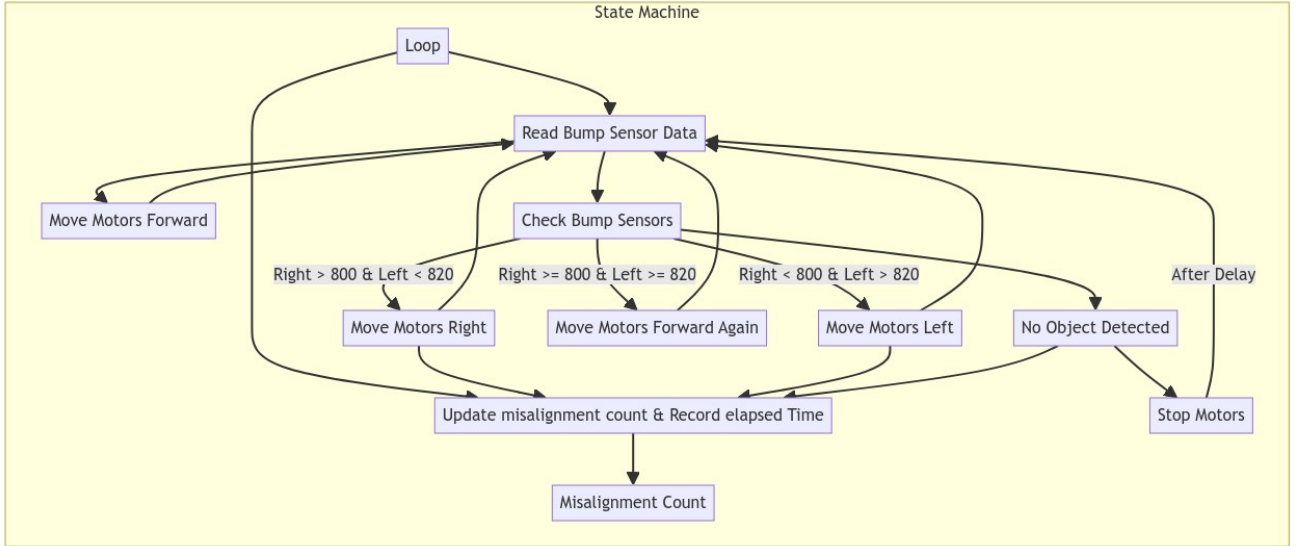


Fig. 5. State machine for the experiment

C. Discussion of Metric(s)

Metrics were chosen to analyse the performance and stability of the robots when pushing the object and compare both configurations in order to determine the orientation with better stability.

- **Misalignment:** The misalignment instances are counted for both robots in each configuration to obtain a plot on the frequency of misalignment of the object from the centre of the bumper of each robot. The misalignment of the robots are indicated with the activation of the bump sensor.
- **Elapsed time:** The elapsed time values are determined at each occurrence of a misalignment. This provides rate of change of misalignment per second to measure the stability of the configuration..
- **Misalignment (Reverse Robot):** In Configuration- 2, the misalignment of the reverse robot is counted and analysed separately. This is due to the fact that it is not exerting effort on to the object like the forward robot.
- **Rate of change of misalignment:** Since the experiment is done to cover a fixed distance of 80 cm using both configurations, this metric provides a better insight of stability in both orientations which is further elaborated in Results section with supporting evidences.

To get the better insight of **stable configuration**, we calculated the rate of change of misalignment as per equation (4).

$$\text{rate_of_change_misalignment} = \frac{\text{misalignment}_t - \text{misalignment}_{(t-1)}}{\text{elapsed_time}} \quad (4)$$

where:

misalignment_t is the number of misalignment's at time t , $\text{misalignment}_{(t-1)}$ is the number of misalignment's at time $t-1$.

Initial plans were to use data gathered from the accelerometer sensor which was expected to provide better understanding of the forces exerted on the moving object

under observation. The data obtained did not provide much evidence in supporting stability of both the configurations.

- **Analysis of metrics:** The metrics obtained through the experiments for the target distance of 80 cm were then analyzed as follows:

Configuration- 1: 10 experimental trials were conducted using two robots in each trial. The average misalignment performance of both robots are calculated for each trial. Subsequently, the overall average performance across all 10 trials are computed by averaging the individual trial averages. This approach provided an aggregated representation of the combined performance of the two robots across the entire set of trials. This was done due to the fact that both the robots exerted effort on to the object when pushing it.

Configuration- 2: A slightly distinctive approach was followed for this configuration as only one of the robots (i.e., forward robot) exerts effort on to the object while the backward robot supports the forward robot in maintaining the orientation of the object. Therefore, for this configuration the average misalignment of each trial data of the 10 trials from the forward robot alone were used for comparison with Configuration- 1. This is done to ensure equality in evaluating the performance of the two configurations. Also, a comparison of the average trial data obtained from both the forward and reverse robots of Configuration- 2 were then plotted and studied independently.

IV. RESULTS

In **Fig. 6** the graph of misalignment count over elapsed time shows the relationship between the time taken for the robots to lose contact with the object and the number of times they deviated from their desired alignment of the centre of the bump sensor. The graph has two plots, one for Configuration- 1 and one for Configuration- 2. The following can be inferred from the graph:

- 1) **Configuration- 1 has higher misalignment count:** The points for Configuration- 1 are generally higher than the points for Configuration- 2, indicating that the robots in Configuration- 1 experienced more misalignment errors than the robots in Configuration- 2. This suggests that Configuration- 1 is more challenging for maintaining proper alignment, possibly due to the increased complexity of coordinating movements.
- 2) **Configuration- 1 has lower elapsed time:** The points for Configuration- 1 are steeper and more close to the y- axis whereas the the points for the Configuration- 2 are flatter and more close to the x- axis. This shows that although Configuration- 1 takes less time (i.e., faster) to complete the objective of pushing the object to the target distance, it has higher misalignment count compared to Configuration- 2. This suggests that Configuration- 2 is more efficient and stable in pushing the object than Configuration- 1 but slower than Configuration- 1.
- 3) **Positive correlation between elapsed time and misalignment count:** The points for both configurations show an upward trend, indicating that as the elapsed time increases, the misalignment count also increases. This implies that the longer the robots push the object, the more likely they are to encounter errors in their positioning and coordination.

Fig. 7 shows the relationship between the elapsed time and the rate of change of misalignment count for the two configurations of robots. The following can be inferred from the graph:

- 1) **Configuration- 1 has higher rate of change of misalignment** than Configuration- 2, this shows that the robots in Configuration- 1 lose contact with the object more frequently than the robots in Configuration- 2.
 - 2) **The rate of change of misalignment decreases over time for both configurations**, this shows that the robots become more stable and aligned as the objects are pushed further.
- The rate of change of misalignment is higher for Configuration- 1 than Configuration- 2, which mean that the robots in Configuration- 1 experience more fluctuations in their alignment than the robots in Configuration- 2. This suggests that Configuration- 2 is more stable than Configuration- 1.

Fig. 8 shows the relationship between misalignment count vs elapsed time between the forward and reverse robots of Configuration- 2. The following can be inferred from the graph:

- 1) **Forward robot:** The graph shows that the forward robot had a lower misalignment count than the reverse robot, indicating that it had more control over the object's orientation. The misalignment count also remained relatively stable over time.
- 2) **Reverse robot:** The graph shows that the reverse robot had a higher misalignment count than the forward robot, indicating that it had more difficulty in maintaining a stable contact with the object. The misalignment count also increased over time, suggesting that the reverse robot faced more challenges

in maintaining the orientation of the object with the forward robot.

Rate of change of misalignment between forward and the reverse robot, can be inferred from the slope of the plots in Fig. 8. The constant slope of the plots proves that the rate of change of misalignment of both robots were uniform through out the experiment.



Fig. 6. Misalignment vs Elapsed time between Configuration- 1 and Configuration- 2

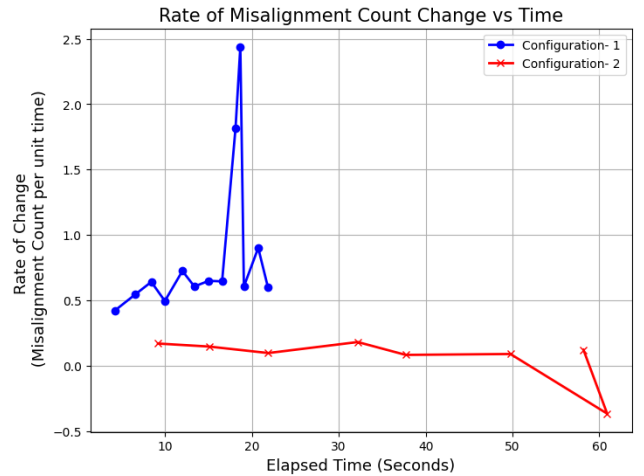


Fig. 7. Rate of change misalignment vs Elapsed time for Configuration- 1 and Configuration- 2

V. DISCUSSION

H1 proposed that Configuration- 1 will be effective in terms of time as both robots are acting on the same side of the box. This is proven by the data as (shown in Fig. 6), Configuration- 1 took lower time of 24 secs to travel target distance (80 cm) whereas Configuration- 2 took 68 secs to travel the same distance. Thus, Configuration- 1 is 2.83 times faster than Configuration- 2.

H2 proposed that Configuration- 2 will be more stable due to the counter forces which will maintain minimum deviation. It is implied from Fig. 7 that the rate of misalignment in Configuration- 2 is relatively low as compared to Configuration- 1. The data insights show us that even

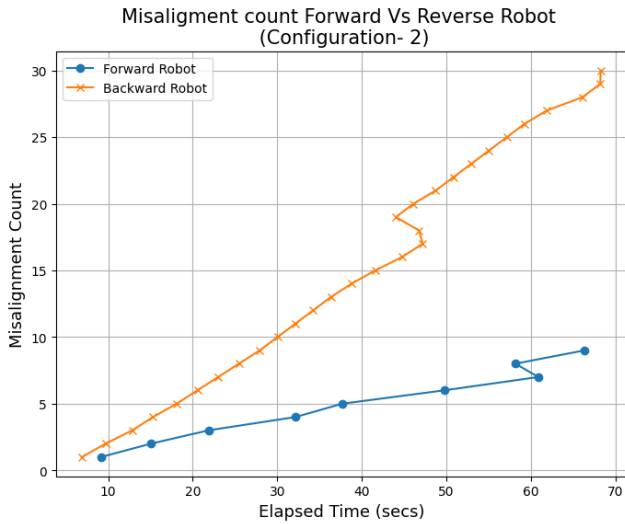


Fig. 8. Misalignment count for forward and backward robots for Configuration- 2

though the configuration 2 took longer time, Configuration- 2 is more stable than Configuration- 1 in terms of misalignment. This is evident from the graph that, Configuration- 2 experiences less change in misalignment over time compared to Configuration- 1. This is further supported by the fact that the rate of change of misalignment for Configuration- 2 is consistently lower than Configuration- 1 indicating low variations.

The experiment treats the two robots as independent entities, without considering their interactions and coordination in order to coordinate their responses to the changing environment. Also this experiment primarily focuses on the stability (maintaining contact between robots and the object for a long duration), which is a single metric of performance. Additional metrics such as trajectory error, force and energy consumption could provide a more comprehensive assessment of the configurations.

Based on the results obtained from the experiment it can be determined that, Configuration- 1 would be better suited for applications where speed plays an important role, such as logistics and warehouse operations to precisely position and package items and Configuration- 2 would be better suited for applications where precise manipulations are required such as assembly lines in the manufacturing sector.

The absence of a PID control limits accurate movement of the robots. The use of encoders would have provided precise distance travelled data. This data coupled with misalignment count could have provided better insight about the stability and performance in comparing each robot in both configuration.

VI. CONCLUSION

Based on the analysis of the misalignment rate, it can be concluded that Configuration- 2 exhibits greater stability in terms of misalignment prevention and correction compared to configuration 1. This is evident from the lower misalignment rate for configuration 2. This also suggests that the opposing orientation by the two robots in Configuration- 2 support against any deviations from the

desired position, making it more resilient to misalignment. These findings highlight the benefit of employing this orientation in the design of collaborative robotic systems for tasks that require precise alignment and manipulation.

VII. FURTHER WORK

For further comprehensive assessments of orientation and manipulation, both configurations could be tested with varying box shapes (rectangular, cylindrical, spherical).

An enhanced feedback loop can be used to iteratively improve experimental setup, robot control, and methodology based on trial insights.

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